1 Planning (15 marks)

Defining the problem (3 marks)

P *A* is the independent variable and *V* is the dependent variable or vary *A* and measure *V*. [1]
P Keep the number of turns on coil Y or coil X constant. [1]
P Keep the current in coil X constant. [1]

Methods of data collection (5 marks)

M1 Two independent coils labelled X and Y; coil Y wound over coil X. [1]
M2 Alternating power supply/signal generator connected to coil X. [1]
M3 Coil Y connected to voltmeter/c.r.o. in a workable circuit. [1]
M4 Measure diameter/radius/lengths with a ruler/vernier callipers. [1]
M5 Method to determine area. [1]

Method of analysis (2 marks)

A Plot a graph of *V* against *A*. [1]
A Relationship valid if straight line through origin. [1]

Safety considerations (1 mark)

S Precaution linked to (large) current in coil/heating, e.g. switch off when not in use to avoid overheating coil; do not touch coil because it is hot. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use large current in coil X/large number of turns/high frequency a.c. to produce measurable e.m.f.
2 Detail on measuring e.m.f., e.g. height × *y*-gain on CRO.
3 Keep frequency of power supply constant.
4 Use of rheostat to keep current constant in coil X.
5 Monitor with a.c. ammeter.
6 Avoid other alternating magnetic fields.
7 Repeat measurement for *r* or *d* or lengths and average.

Do not allow vague computer methods.

[Total: 15]

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